

From Launch Files to Manifests

Launch File (control.launch.xml)

```
<arg name="launch_validator"/>

<push-ros-namespace ns="/control"/>

<node pkg="controller"/>
  <remap from="~/cmd" to="/control/cmd"/>

<group if="$(var launch_validator)">
  <node pkg="validator"/>
  <remap from="~/input" to="/control/cmd"/>
</group>

<include file="system.launch.xml"/>
```

Declares *what to run* and *how to wire names*.
Topics, types, QoS, and rates are implicit in source code.

Manifest (control.yaml)

```
args:
  launch_validator:
    type: bool

nodes:
  controller:
    pub: [cmd]
  validator:
    if: $(var launch_validator)
    sub: [cmd]

topics:
  control_cmd:
    type: Control
    pub: [controller/cmd]
    sub: [validator/cmd]
    rate_hz: 30

includes:
  system:
    manifest: system.yaml
```

Declares *what communicates* and *at what quality*.
Topics are first-class with type, QoS, rate, latency.

Launch XML	Manifest YAML	What the manifest adds
<node>	nodes:	Named endpoints (pub/sub/srv/cli)
<remap>	topics:	First-class wiring: type + QoS + rate
<include>	includes:	Child scope with its own manifest
<arg> / <let>	args:	Typed params: bool, choices, string
if="\$(var ...)"	if: / unless:	Conditions on any entity
(in source code)	paths:	Timing contracts: latency, age, drops

Launch XML	Manifest YAML	What the manifest adds
<i>(in source code)</i>	pub: / sub:	Scope boundary interface